



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Parth Manoj Gajare
Roll No. :	74
Experiment No.:	2
Title of the Experiment:	Apply Descriptive Statistics and Data Visualization Techniques for Exploratory Data Analysis
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature:

Date :

2015111: Statistics for Machine Learning and Data Science Lab



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Aim: To apply descriptive statistical measures and data visualization techniques to summarize, analyze, and interpret the distribution and relationships present in the Pima Indians Diabetes Dataset.

Objectives

After completing this experiment, the student will be able to:

- Apply descriptive statistical and visualization techniques to explore the characteristics of a real-world healthcare dataset.
- Interpret distributions, variability, relationships, and patterns obtained from statistical analysis and visualizations.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. Matplotlib
6. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for female patients and is used to study factors associated with diabetes.

The dataset contains 768 observations and 8 input attributes, along with a binary outcome variable indicating diabetes status.

Attribute	Description



Pregnancies	Number of pregnancies
Glucose	Plasma glucose concentration
BloodPressure	Diastolic blood pressure
SkinThickness	Triceps skin fold thickness
Insulin	2-Hour serum insulin
BMI	Body Mass Index
DiabetesPedigreeFunction	Diabetes pedigree function
Age	Age in years
Outcome	1 = Diabetic, 0 = Non-diabetic

Procedure

1. Load the Pima Indians Diabetes Dataset into a Pandas DataFrame.
 2. Inspect the dataset structure, dimensions, column names, and data types.
 3. Calculate descriptive statistics including mean, median, mode, minimum, maximum, variance, and standard deviation for selected numerical variables.
 4. Analyze the distribution of variables using histograms and boxplots.
 5. Analyze the frequency of the Outcome variable using a bar chart.
 6. Generate scatter plots to study relationships such as Glucose vs. BMI and Age vs. Glucose.
 7. Generate a pair plot for selected numerical variables.
 8. Interpret the statistical summaries and visualizations and record important observations.
- ### Description of the Experiment



Descriptive statistics summarize the important characteristics of a dataset using numerical measures, while data visualization presents these characteristics graphically.

The experiment follows the workflow: Dataset, Descriptive Statistics, Distribution Analysis, Visualization, Relationship Analysis, Interpretation,

Students will use the Pima Diabetes Dataset to analyze the central tendency, variability, distributions, and relationships among health-related variables.

Detailed Description of the Statistical techniques:

Mean, Median, and Mode

These measures describe the central or typical value of a dataset.

$$\bar{x} = \frac{\sum x_i}{n}$$

The median is the middle value, while the mode is the most frequently occurring value.

Range, Variance, and Standard Deviation These

measures describe data variability.

$$\begin{aligned} \text{Range} &= \text{Maximum} - \text{Minimum} \\ s^2 &= \frac{\sum (x_i - \bar{x})^2}{n - 1} \\ s &= \sqrt{s^2} \end{aligned}$$

Histogram

Used to study the distribution, spread, and shape of numerical variables such as Glucose, BMI, and Age.



Boxplot

Used to visualize the median, quartiles, spread, and potential outliers.

$$IQR = Q_3 - Q_1$$

Bar Chart

Used to display the frequency of categorical or binary variables such as Outcome.

Scatter Plot

Used to examine the relationship between two numerical variables, such as:

- Glucose and BMI
- Age and Glucose

Pair Plot

Used to simultaneously visualize pairwise relationships and distributions among multiple numerical variables.

CODE:-

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
sns.set(style="whitegrid")
```

```
df = pd.read_csv("diabetes.csv")
```



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```
print("=" * 60)
print("PIMA INDIANS DIABETES DATASET")
print("=" * 60)
```

```
print("\nFirst 5 Rows")
print(df.head())
```

```
print("\nLast 5 Rows")
print(df.tail())
```

```
print("\nDataset Shape")
print(df.shape)
```

```
print("\nColumn Names")
print(df.columns.tolist())
```

```
print("\nData Types")
print(df.dtypes)
```

```
print("\nDataset Information")
df.info\(\)
```

```
print("\nMissing Values")
print(df.isnull().sum())
```



```
print("\n" + "=" * 60)
print("DESCRIPTIVE STATISTICS")
print("=" * 60)
```

```
print("\nSummary Statistics")
print(df.describe())
```

```
print("\nMean")
print(df.mean(numeric_only=True))
```

```
print("\nMedian")
print(df.median(numeric_only=True))
```

```
print("\nMode")
print(df.mode().iloc[0])
```

```
print("\nMinimum")
print(df.min(numeric_only=True))
```

```
print("\nMaximum")
print(df.max(numeric_only=True))
```

```
print("\nVariance")
print(df.var(numeric_only=True))
```

```
print("\nStandard Deviation")
```



```
print(df.std(numeric_only=True))
```

```
print("\nRange")
```

```
range_values = df.max(numeric_only=True) - df.min(numeric_only=True)
```

```
print(range_values)
```

```
df.hist(figsize=(15, 12), bins=20, edgecolor='black')
```

```
plt.suptitle("Histograms of Numerical Variables", fontsize=16)
```

```
plt.tight_layout()
```

```
plt.show()
```

```
plt.figure(figsize=(14, 7))
```

```
sns.boxplot(data=df)
```

```
plt.xticks(rotation=45)
```

```
plt.title("Boxplots of Numerical Variables")
```

```
plt.show()
```

```
columns = [
```

```
    "Pregnancies",
```

```
    "Glucose",
```

```
    "BloodPressure",
```

```
    "SkinThickness",
```

```
    "Insulin",
```

```
    "BMI",
```

```
    "DiabetesPedigreeFunction",
```

```
    "Age"
```



]

for col in columns:

```
plt.figure(figsize=(5, 5))
sns.boxplot(y=df[col], color="skyblue")
plt.title(f"Boxplot of {col}")
plt.show()
```

```
plt.figure(figsize=(6, 5))
sns.countplot(x="Outcome", data=df)
plt.title("Frequency of Outcome")
plt.xlabel("Outcome (0 = Non-Diabetic, 1 = Diabetic)")
plt.ylabel("Count")
plt.show()
```

```
plt.figure(figsize=(7, 5))
sns.scatterplot(
    x="Glucose",
    y="BMI",
    hue="Outcome",
    data=df
)
```

```
plt.title("Glucose vs BMI")
plt.show()
```



```
plt.figure(figsize=(7, 5))
```

```
sns.scatterplot(
```

```
    x="Age",
```

```
    y="Glucose",
```

```
    hue="Outcome",
```

```
    data=df
```

```
)
```

```
plt.title("Age vs Glucose")
```

```
plt.show()
```

```
pair_columns = [
```

```
    "Glucose",
```

```
    "BMI",
```

```
    "Age",
```

```
    "BloodPressure",
```

```
    "Outcome"
```

```
]
```

```
sns.pairplot(
```

```
    df[pair_columns],
```

```
    hue="Outcome",
```

```
    diag_kind="hist"
```

```
)
```

```
plt.show()
```



```
plt.figure(figsize=(10, 8))

sns.heatmap(
    df.corr(),
    annot=True,
    cmap="coolwarm",
    linewidths=0.5
)

plt.title("Correlation Matrix")
plt.show()

std_values = df.std(numeric_only=True)

highest_std_variable = std_values.idxmax()

print("\nVariable with Highest Standard Deviation")
print(highest_std_variable)
print("Standard Deviation =", std_values.max())

difference = abs(
    df.mean(numeric_only=True) -
    df.median(numeric_only=True)
)

print("\nDifference Between Mean and Median")
```



```
print(difference)

largest_difference = difference.idxmax()

print("\nVariable with Largest Mean-Median Difference")
print(largest_difference)

print("\nOutlier Detection using IQR")

for col in columns:

    Q1 = df[col].quantile(0.25)
    Q3 = df[col].quantile(0.75)

    IQR = Q3 - Q1

    lower_limit = Q1 - 1.5 * IQR
    upper_limit = Q3 + 1.5 * IQR

    outliers = df[
        (df[col] < lower_limit) |
        (df[col] > upper_limit)
    ]

    print(f"{col}: {len(outliers)} outliers")
```



```
print("\n" + "=" * 60)

print("OBSERVATIONS")

print("=" * 60)

print("1. Histograms show the distribution of numerical variables.")

print("2. Boxplots indicate outliers in Insulin, BMI, Pregnancies and DiabetesPedigreeFunction.")

print("3. Glucose has a positive relationship with diabetes outcome.")

print("4. Scatter plots show a weak positive relationship between Glucose and BMI.")

print("5. Correlation heatmap shows Glucose has one of the strongest correlations with Outcome.")

print("\nExperiment Completed Successfully.")
```

OUTPUT:-

```
=====
PIMA INDIANS DIABETES DATASET
=====

First 5 Rows
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
0           6     148           72           35         0   33.6              0.627  50      1
1           1      85           66           29         0   26.6              0.351  31      0
2           8     183           64           0          0   23.3              0.672  32      1
3           1      89           66           23         94   28.1              0.167  21      0
4           0     137           40           35        168   43.1              2.288  33      1

Last 5 Rows
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
763          10     101           76           48        180   32.9              0.171  63      0
764           2     122           70           27         0   36.8              0.340  27      0
765           5     121           72           23        112   26.2              0.245  30      0
766           1     126           60           0          0   30.1              0.349  47      1
767           1      93           70           31         0   30.4              0.315  23      0

Dataset Shape
(768, 9)
```



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```
Column Names
['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome']

Data Types
Pregnancies      int64
Glucose           int64
BloodPressure     int64
SkinThickness     int64
Insulin           int64
BMI               float64
DiabetesPedigreeFunction float64
Age               int64
Outcome           int64
dtype: object

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---
0   Pregnancies            768 non-null   int64
1   Glucose                768 non-null   int64
2   BloodPressure          768 non-null   int64
3   SkinThickness          768 non-null   int64
4   Insulin                768 non-null   int64
5   BMI                    768 non-null   float64
6   DiabetesPedigreeFunction 768 non-null   float64
7   Age                    768 non-null   int64
8   Outcome                768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

```
Missing Values
Pregnancies      0
Glucose           0
BloodPressure     0
SkinThickness     0
Insulin           0
BMI               0
DiabetesPedigreeFunction 0
Age               0
Outcome           0
dtype: int64

=====
DESCRIPTIVE STATISTICS
=====

Summary Statistics
Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin  BMI  DiabetesPedigreeFunction  Age  Outcome
count  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000  768.000000
mean    3.845052  120.894531  69.105469  20.536458  79.799479  31.992578  0.471876  33.240885  0.348958
std     3.369578  31.972618  19.355807  15.952218  115.244002  7.884160  0.331329  11.760232  0.476951
min     0.000000  0.000000  0.000000  0.000000  0.000000  0.000000  0.078000  21.000000  0.000000
25%     1.000000  99.000000  62.000000  0.000000  0.000000  27.300000  0.243750  24.000000  0.000000
50%     3.000000  117.000000  72.000000  23.000000  30.500000  32.000000  0.372500  29.000000  0.000000
75%     6.000000  140.250000  80.000000  32.000000  127.250000  36.600000  0.626250  41.000000  1.000000
max    17.000000  199.000000  122.000000  99.000000  846.000000  67.100000  2.420000  81.000000  1.000000
```



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```
Mean
Pregnancies      3.845052
Glucose          120.894531
BloodPressure    69.105469
SkinThickness    20.536458
Insulin          79.799479
BMI              31.992578
DiabetesPedigreeFunction  0.471876
Age              33.240885
Outcome          0.348958
dtype: float64
```

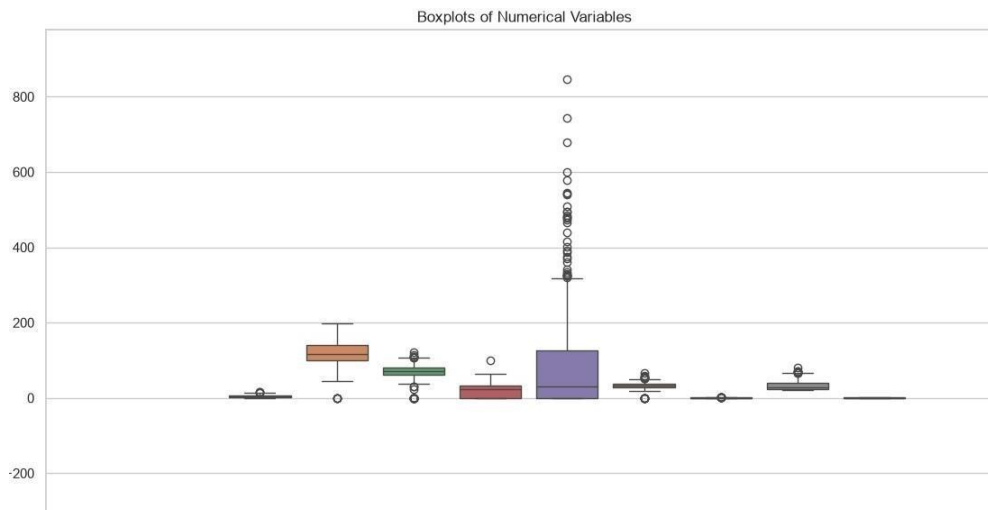
```
Median
Pregnancies      3.0000
Glucose          117.0000
BloodPressure    72.0000
SkinThickness    23.0000
Insulin          30.5000
BMI              32.0000
DiabetesPedigreeFunction  0.3725
Age              29.0000
Outcome          0.0000
dtype: float64
```

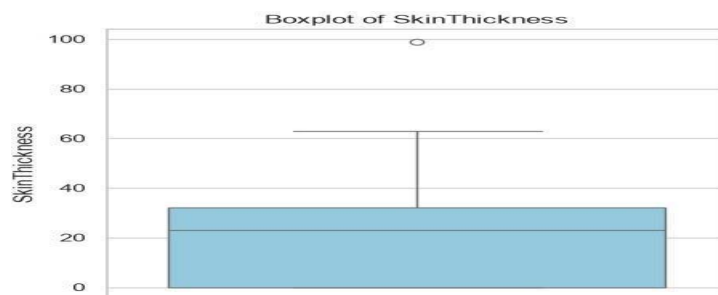
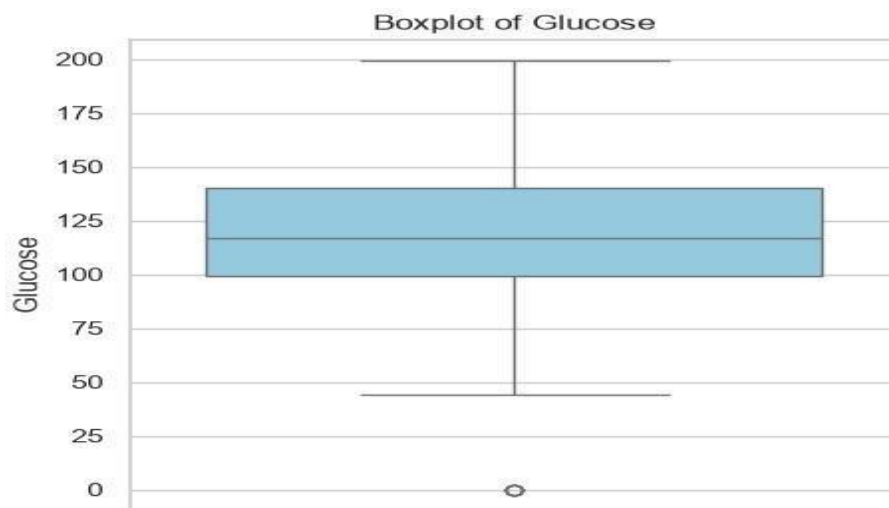
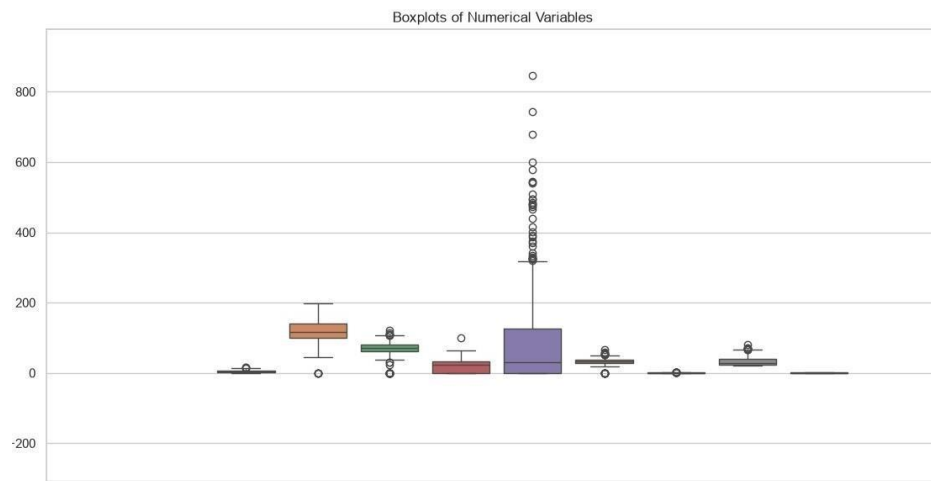
```
Mode
Pregnancies      1.000
Glucose          99.000
BloodPressure    70.000
SkinThickness    0.000
Insulin          0.000
BMI              32.000
DiabetesPedigreeFunction  0.254
Age              22.000
Outcome          0.000
Name: 0, dtype: float64
```

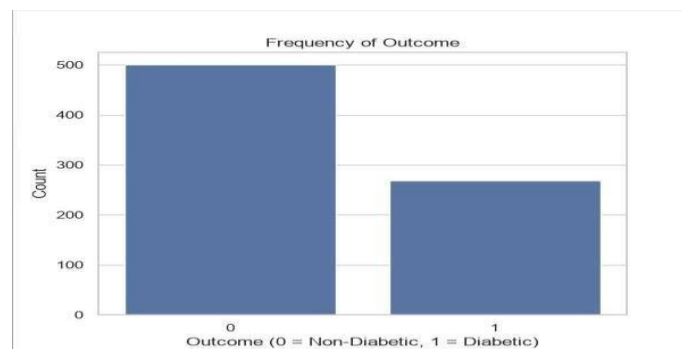
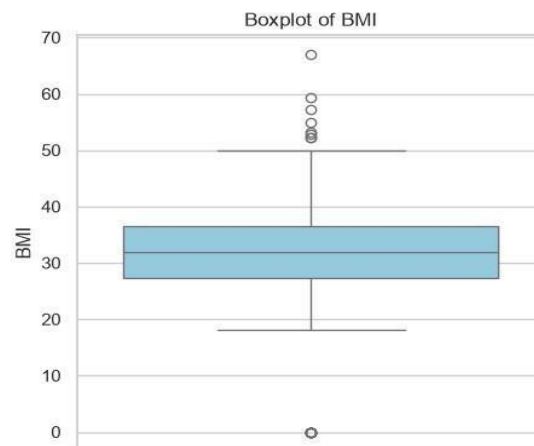
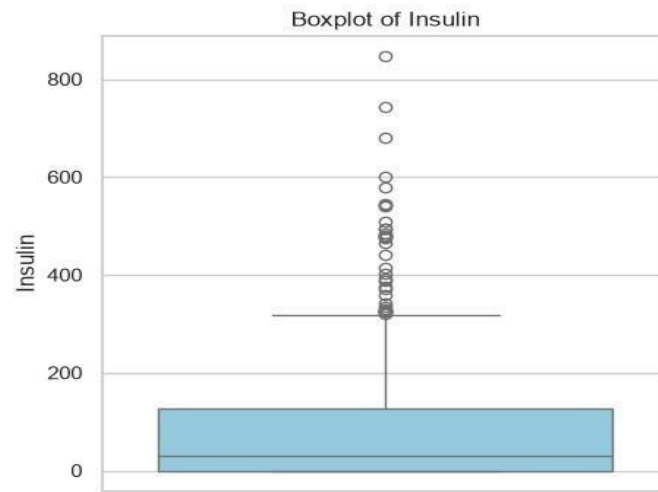
```
Minimum
Pregnancies      0.000
Glucose          0.000
BloodPressure    0.000
SkinThickness    0.000
Insulin          0.000
BMI              0.000
DiabetesPedigreeFunction  0.078
Age              21.000
Outcome          0.000
dtype: float64
```

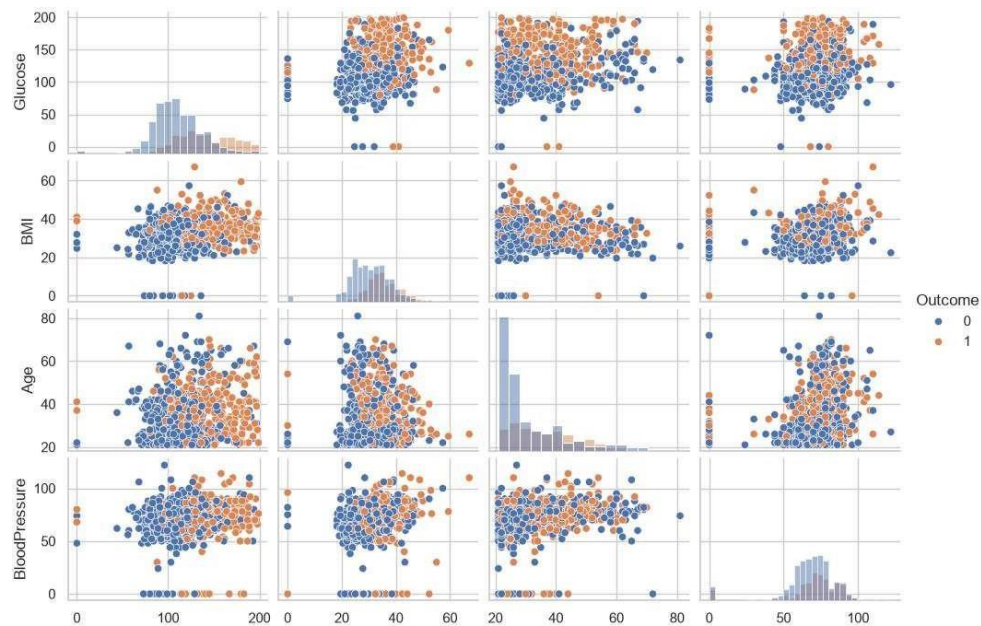
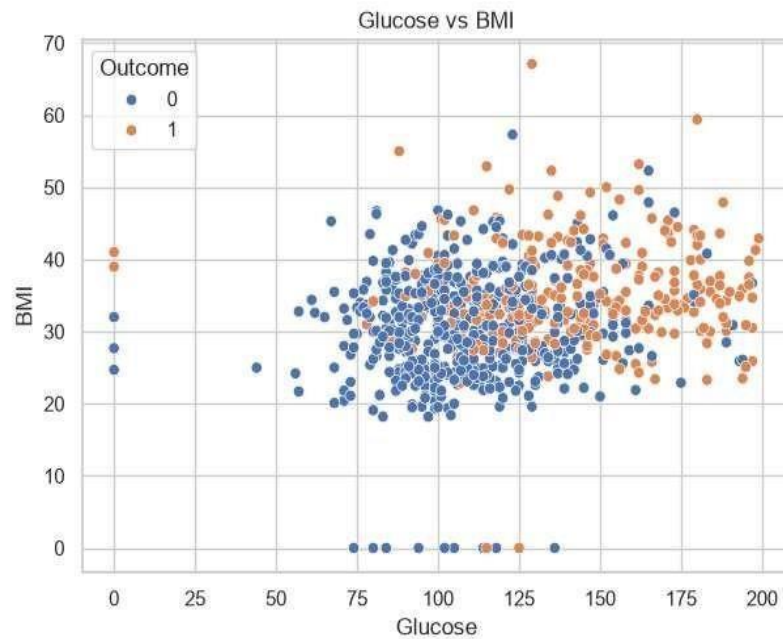
```
Maximum
Pregnancies      17.00
Glucose          199.00
BloodPressure    122.00
SkinThickness    99.00
Insulin          846.00
BMI              67.10
DiabetesPedigreeFunction  2.42
Age              81.00
Outcome          1.00
dtype: float64
```

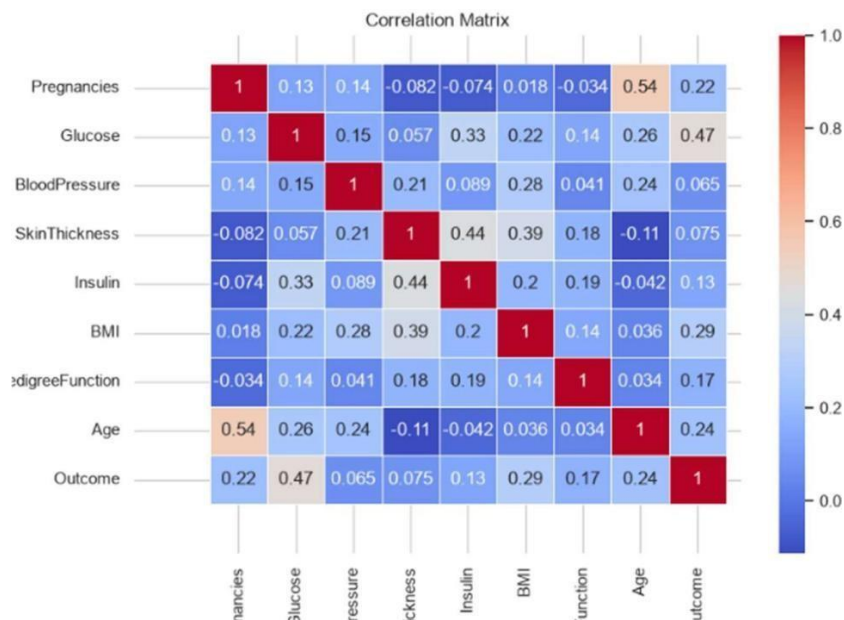
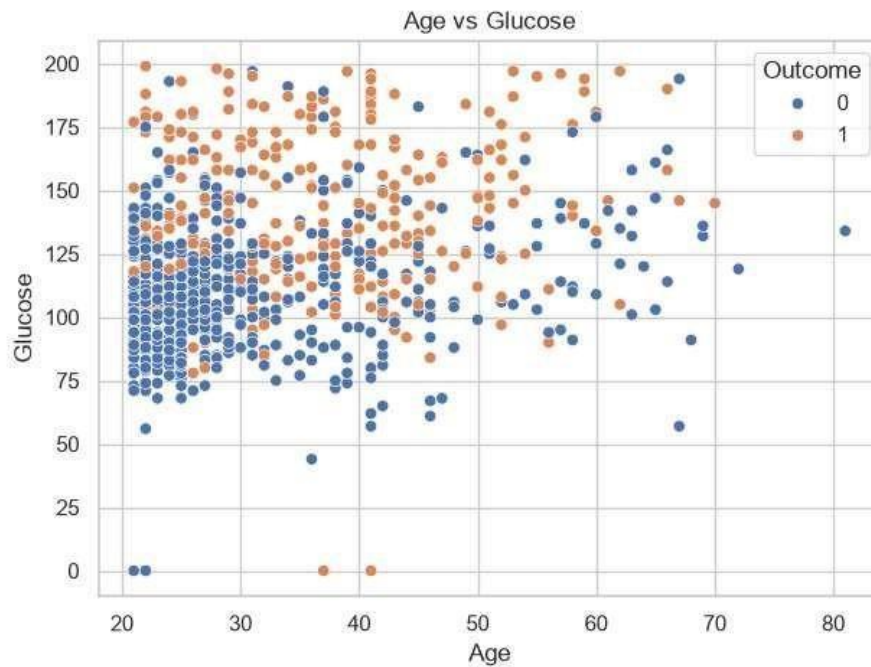
```
Variance
Pregnancies      11.354056
Glucose          1022.248314
BloodPressure    374.647271
SkinThickness    254.473245
Insulin          13281.180078
BMI              62.159984
DiabetesPedigreeFunction  0.109779
Age              138.303046
Outcome          0.227483
dtype: float64
```













Conclusion

Descriptive statistical measures and visualization techniques were applied to the Pima Indians Diabetes Dataset to understand the distribution, variability, and relationships among the variables.

Questions to be Answered by Students

Based on the standard **Pima Indians Diabetes Dataset (768 records)** and the exploratory analysis performed, the answers are:

1. Which numerical variable has the highest standard deviation?

The Insulin variable has the highest standard deviation because its values are spread over a very wide range, with several extremely high values (outliers).

2. Which variable shows the greatest difference between its mean and median?

The Insulin variable shows the largest difference between its mean and median. This occurs because the distribution is highly right-skewed, with many zero values (representing missing/invalid measurements) and a few very large insulin values that increase the mean.

3. What relationship can be observed between Glucose and BMI from the scatter plot?

The scatter plot indicates a weak to moderate positive relationship between Glucose and BMI. Patients with higher BMI tend to have higher glucose levels, although the relationship is not perfectly linear and there is considerable variability.

4. State two important insights obtained from the analysis.

1. Glucose is the strongest indicator of diabetes, with diabetic patients generally having significantly higher glucose levels than non-diabetic patients.
2. Several medical variables (Glucose, BloodPressure, SkinThickness, Insulin, and BMI) contain zero values, which likely represent missing or invalid measurements and should be handled before building predictive models.

5. Which variable(s), if any, contain potential outliers? How were these identified?

Potential outliers were found in the following variables:

- Insulin
- BMI



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- DiabetesPedigreeFunction
- Pregnancies
- (Some analyses also identify a few outliers in SkinThickness and Age.)

These outliers were identified using:

- Boxplots, which visually highlight observations beyond the whiskers.
- The Interquartile Range (IQR) method, where values less than $Q1 - 1.5 \times IQR$ or greater than $Q3 + 1.5 \times IQR$ are considered potential outliers.

6. State two important insights obtained from your exploratory statistical analysis of the dataset.

1. The dataset contains 768 observations, with 8 numerical predictor variables and 1 binary outcome variable, making it suitable for diabetes prediction and classification tasks.
2. The exploratory analysis revealed invalid zero values and several outliers, indicating that data preprocessing (handling missing values and evaluating outliers) is an essential step before applying statistical or machine learning models.